

1. Advanced SQL DATE Functions

1. ADD 6 months to each hire-date

```
SELECT emp-id, name, hire-date,  
       DATEADD(MONTH, 6, hire-date) AS  
       hire-plus-6-months  
FROM Employees;
```

emp-id	name	hire-date	hire-plus-6-months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

2. Age in days from dob to today

```
SELECT student-id, name,  
       DATEDIFF(DAY, dob, CURRENT_DATE) AS  
       age-in-days  
FROM Students;
```

Student-id	name	age-in-days
101	Maya	7585
102	Ethan	7887
103	Sienna	7368

3. Days remaining until each event

```
SELECT event-id, event-name,  
       DATEDIFF(DAY, CURRENT_DATE, event-date) AS  
       days-remaining FROM Events;
```

event-id	event-name	days-remaining
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1	Seminar	- 701
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2	Workshop	- 258
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3	Hackathon	- 482
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4. Days Between issue_date and due_date

```
SELECT invoice_id, issue_date, due_date,
       DATEDIFF (DAY, issue_date, due_date) AS
       days-between
FROM Invoices;
```

invoice-id	issue-date	due-date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

5. Format start-date as 'Month YYYY'

```
SELECT course_id, name,
       TO_CHAR (start_date, 'Month YYYY') AS
       formatted_date
FROM Courses;
```

Course-id	name	formatted-date
201	SQL Basics	may 2025
202	Python	June 2025

6. Build full date from parts

```
SELECT member-id,  
       DATE-FROM-PARTS (start-year,  
                        start-month, start-day) AS full-start-date  
FROM Memberships;
```

member-id	full-start-date
1	2023-05-10
2	2022-11-25

7. Extend renewal-date by 1 year

```
SELECT sub-id, plan,  
       DATEADD(YEAR, 1, renewal-date) AS  
       extended-renewal-date  
FROM Subscriptions;
```

Sub-id	plan	extended-renewal-date
11	Basic	2026-01-01
12	Premium	2026-03-15

8. Current date and days since order

```
SELECT order-id, order-date,  
       CURRENT-DATE AS today-date,  
       DATEDIFF(DAY, order-date, CURRENT-DATE)  
       AS days-since-order  
FROM Orders;
```

order-id	order-date	today-date	days-since-order
1001	2025-04-15	2026-05-17	397
1002	2025-04-10	2026-05-17	402

9. Extract year from training-date

```
SELECT training-id, topic,
       Extract (YEAR FROM training-date) AS
       training-year
FROM Trainings;
```

training-id	topic	training-year
1	Safety	2025
2	Compliance	2025

10. Extract hour and minute from published-on

```
SELECT post-id, title,
       Extract (Hour FROM published-on) AS
       hour-published,
       Extract (Minute FROM published-on) AS
       minute-published;
```


10. Extract hour and minute from published-on

```
SELECT post-id, title,  
       EXTRACT (HOUR FROM published-on) AS  
       hour-published,  
       EXTRACT (MINUTE FROM published-on) AS  
       minute-published  
FROM Blog-posts;
```

post-id	title	hour-published	minute-published
1	SQL Tips	10	45
2	Data Cleaning	16	45

11. Days left until license expiry

```
SELECT driver-id, license-expiry,  
       DATEDIFF (DAY, CURRENT-DATE,  
       license-expiry) AS days-left  
FROM Drivers;
```

driver-id	license-expiry	days-left
301	2025-08-10	-280
302	2023-12-31	-868

12. Seconds since message was sent

```
SELECT message-id, sent-timestamp,  
       CURRENT-TIMESTAMP AS  
       CURRENT-
```

12. Seconds since message was sent

```
SELECT message-id, sent-timestamp,  
       CURRENT-TIMESTAMP AS  
       current-timestamp,  
       DATEDIFF(SECOND, sent-timestamp,  
CURRENT-TIMESTAMP) AS seconds-since-sent  
FROM Messages;
```

message-id	sent-timestamp	current-timestamp	seconds-since-sent
1	2025-04-17 09:32:45	2026-05-17 00:00:00	84,238,835
2	2025-04-18 23:59:55	2026-05-17 00:00:00	34,300,801

13. Add 15 days to return-date

```
SELECT return-id, return-date,  
       DATEADD(DAY, 15, return-date) AS  
       restock-date  
FROM Returns;
```

return-id	return-date	restock-date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

14. Convert string to date

```
SELECT assign-id  
       TO-DATE (assigned-on 'YY-YY-MM-DD') AS  
       assigned-on-date  
FROM assignments;
```

assign-id	assigned-on-date
1	2025-03-01
2	2025-03-05

15. Format scheduled-time as readable string

```
SELECT meeting-id,  
       TO-CHAR (scheduled-time, 'FM Month FM DD,  
       YYYY "at" FM HH:MI AM') AS  
       formatted-meeting-time  
FROM Meetings;
```

meeting-id	formatted-meeting-time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 AM.